Capture the Flag 7: Stop Putting Everything on One Screen

Goal: Split your Student Directory into multiple screens using Flutter navigation, while using a modal for editing student information.

Instructor Note
These activities require your own problem-solving and research skills.
DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🏴 Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding multiple screens.

Create:

`feature/multi-screen-navigation`

Screenshot:

- Terminal showing your current branch

```
PS C:\Users\admin\Documents\FlutterProjects\Multi-Screen-Navigation> git checkout -b feature/multi-screen-navigation
Switched to a new branch 'feature/multi-screen-navigation'
PS C:\Users\admin\Documents\FlutterProjects\Multi-Screen-Navigation>
```

🏴 Flag 1 — Create the Student List Screen

Objective: Move your Student Directory into its own screen.

Create a dedicated **Student List Screen** containing your existing student list.

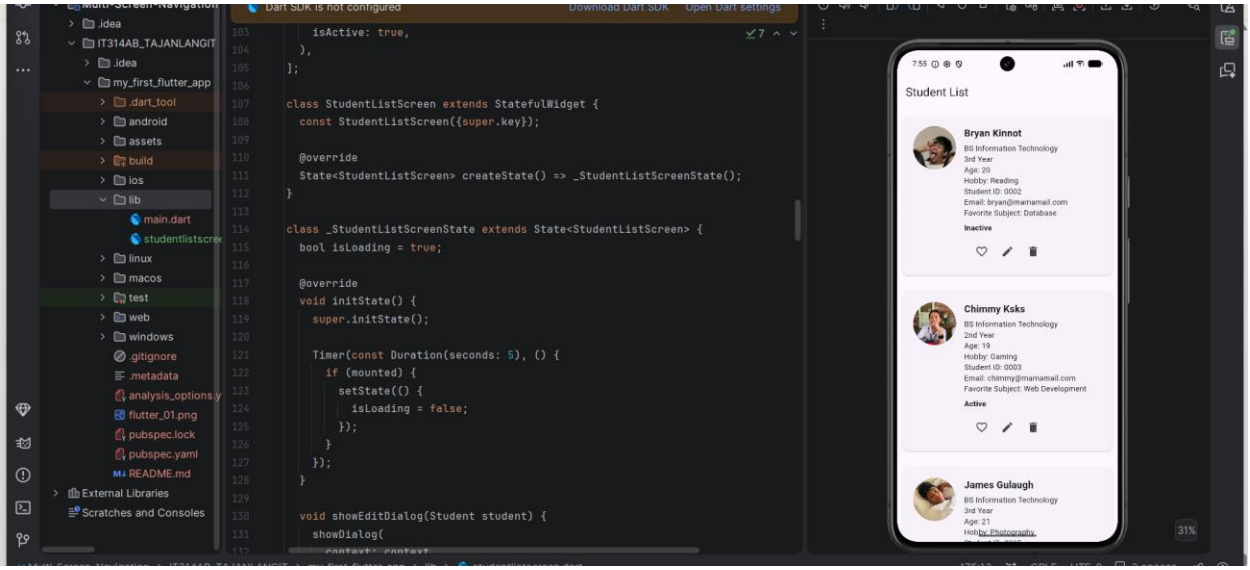
Research how Flutter creates separate screens using `Widget` classes.

Test

The Student Directory should function as its own screen.

Screenshot:

- Student List Screen
- Student List screen code
- Application showing the screen



🏴 Flag 2 — Create the Student Details Screen

Objective: Create a separate screen for viewing a student's information.

When a student is selected, navigate:

Student List → Student Details

Research Flutter's `Navigator` and how to move between screens.

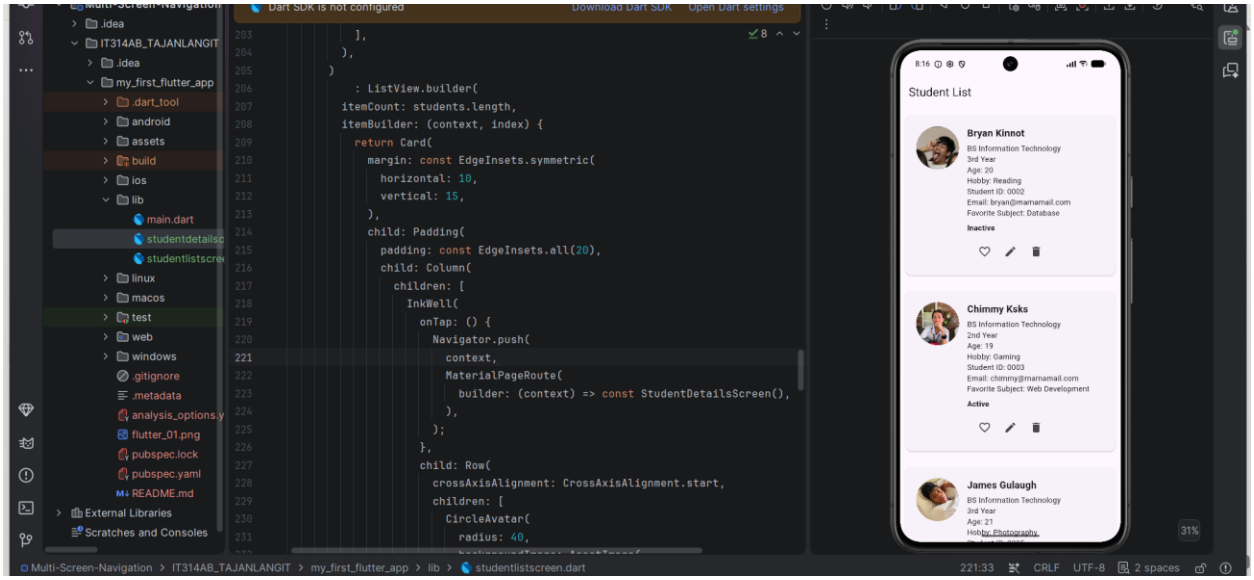
Test

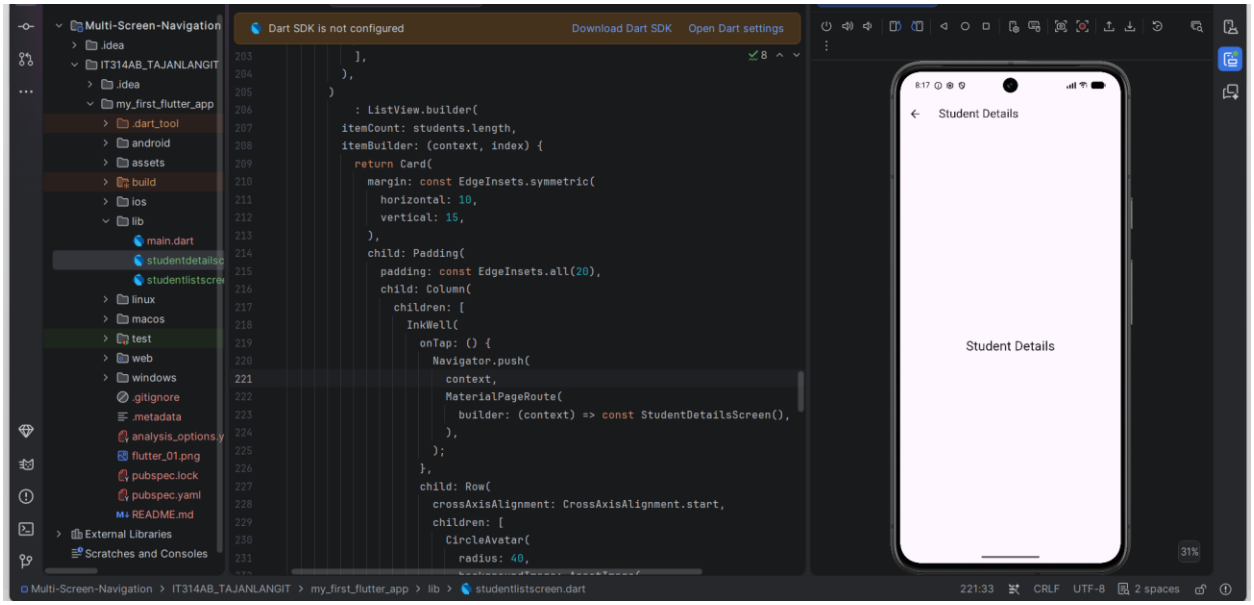
1. Select a student.
2. Open the Student Details Screen. (No data displayed yet)
3. Return to the Student List using back navigation.

Screenshot:

- Student List Screen

- Student Details Screen
- Navigation code





🏴 Flag 3 — Pass Student Data

Objective: Pass the selected student's data to the Student Details Screen.

The Details Screen should display the correct information for the selected student.

Display relevant information such as:

- Name
- Student ID
- Course
- Status
- Other existing student data

Research how data can be passed between Flutter screens.

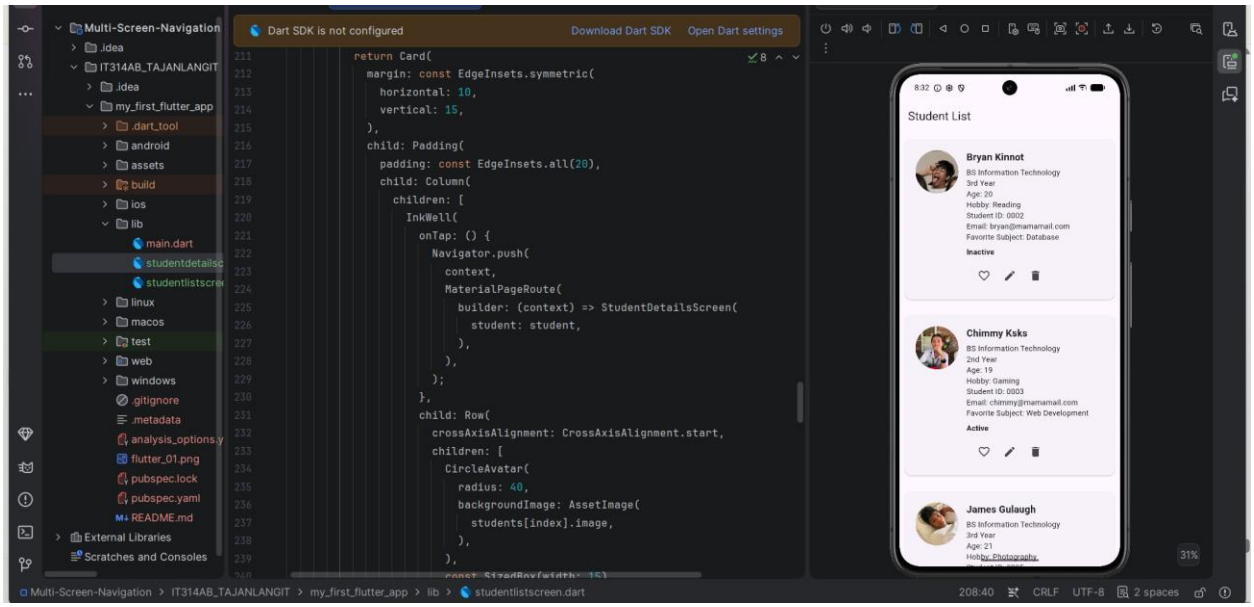
Test

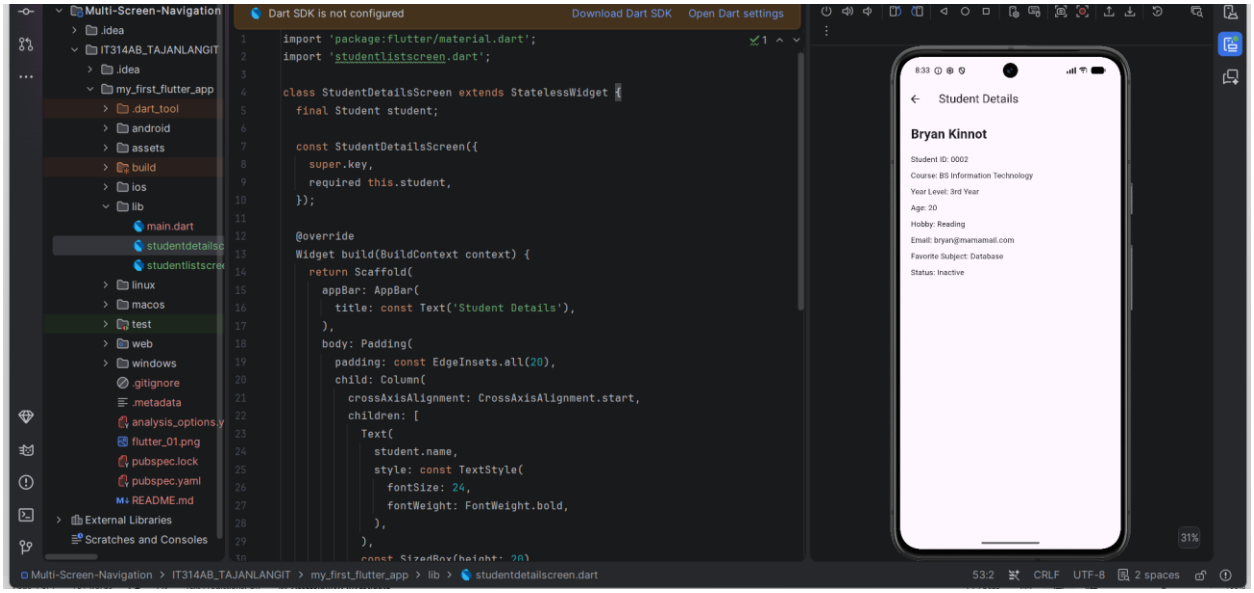
Selecting different students should display their corresponding information.

Student A → Student A's Details
Student B → Student B's Details

Screenshot:

- Selected student
- Student Details
- Data-passing code





🏴 Flag 4 — Edit with a Modal

Objective: Allow administrators to edit student information without creating another screen.

Add an **Edit** action to the Student Details Screen.

When selected, display an editing interface using a **modal**, such as a dialog or bottom sheet.

Research Flutter's:

- `showDialog`
- `showModalBottomSheet`

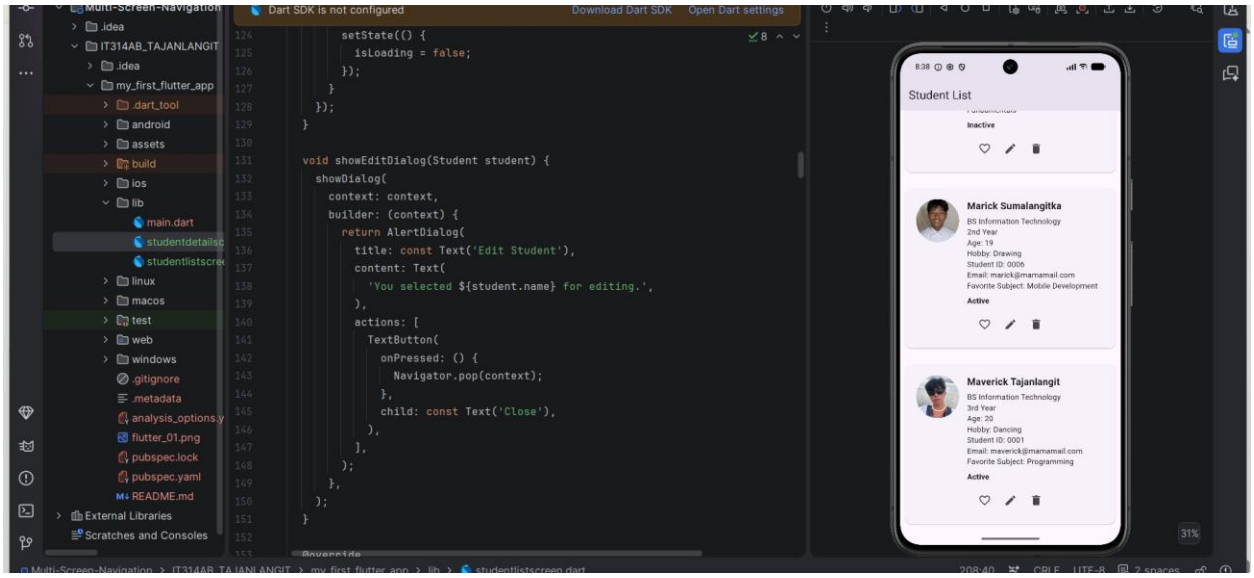
You may choose whichever approach is appropriate for your design.

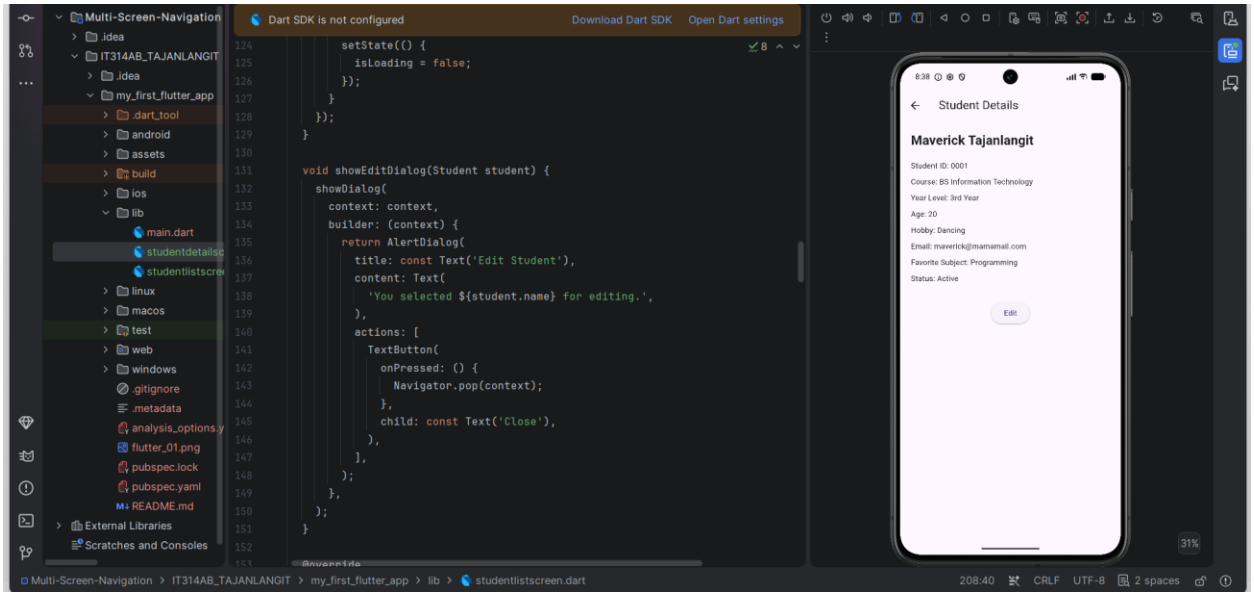
Test

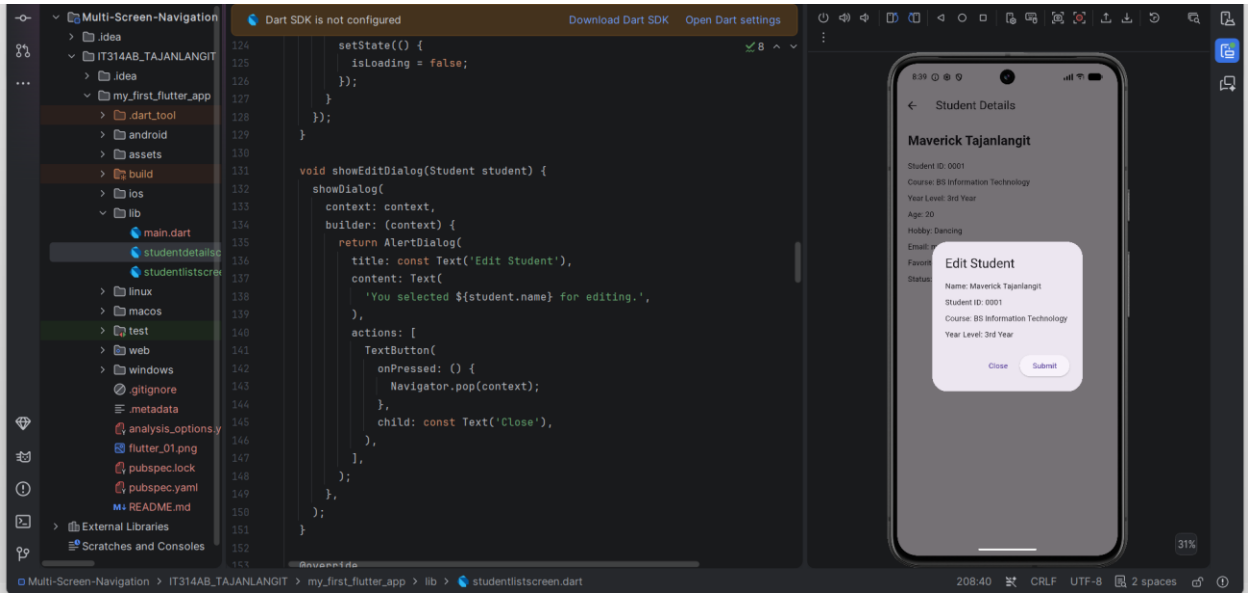
1. Open Student Details.
2. Select Edit.
3. An editing modal should appear.
4. The modal should display the appropriate student's information.
5. Close or submit the modal.

Screenshot:

- Student Details Screen
- Edit modal
- Modal code


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analysis_options.y
flutter_01.png
pubspec.lock
pubspec.yaml
README.md
External Libraries
Scratches and Consoles
setState(() {
isLoading = false;
});
}
});
}
void showEditDialog(Student student) {
showDialog(
context: context,
builder: (context) {
return AlertDialog(
title: const Text('Edit Student'),
content: Text(
'You selected ${student.name} for editing.',
),
actions: [
TextButton(
onPressed: () {
Navigator.pop(context);
},
child: const Text('Close'),
),
],
);
},
);
}
@override
Student List
Inactive
Marick Sumalangitka
BS Information Technology
2nd Year
Age: 19
Hobby: Drawing
Student ID: 0006
Email: marick@mamamail.com
Favorite Subject: Mobile Development
Active
Maverick Tajanlangit
BS Information Technology
3rd Year
Age: 20
Hobby: Dancing
Student ID: 0001
Email: maverick@mamamail.com
Favorite Subject: Programming
Active
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setState(() {
isLoading = false;
});
}
});
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void showEditDialog(Student student) {
showDialog(
context: context,
builder: (context) {
return AlertDialog(
title: const Text('Edit Student'),
content: Text(
'You selected ${student.name} for editing.',
),
actions: [
TextButton(
onPressed: () {
Navigator.pop(context);
},
child: const Text('Close'),
),
],
);
},
);
}
@override
Student Details
Maverick Tajanlangit
Student ID: 0001
Course: BS Information Technology
Year Level: 3rd Year
Age: 20
Hobby: Dancing
Email: maverick@mamamail.com
Favorite Subject: Programming
Status: Active
Edit
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🏴 Flag 5 — Create the Add Student Screen

Objective: Create a separate screen for adding a new student.

Add an action from the Student List that navigates to:

Student List → Add Student

Create the **Add Student Screen**.

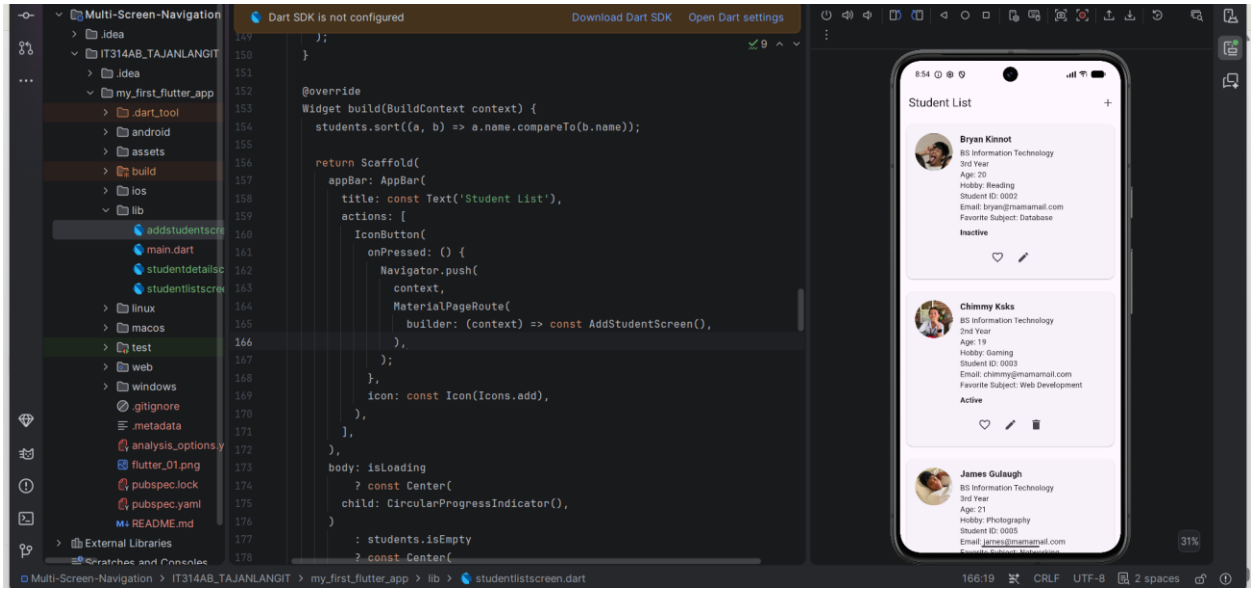
For now, focus on creating the screen and navigation structure.

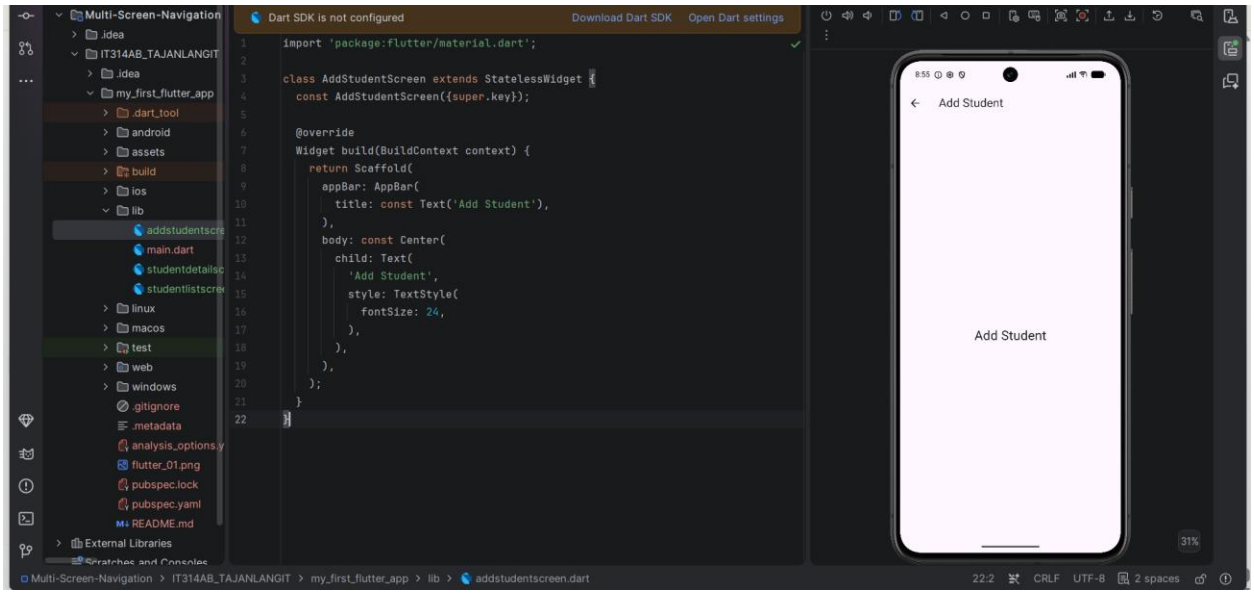
Test

Selecting Add Student should open the Add Student Screen.

Screenshot:

- Student List with Add action
- Add Student Screen
- Navigation code





🏴 Flag 6 — Prepare the Add Student Screen

Objective: Prepare the Add Student Screen for future functionality.

Organize the screen so it is ready to receive student input.

Do not implement the student input fields yet.

The screen should simply exist as a separate destination.

Your application should now have:

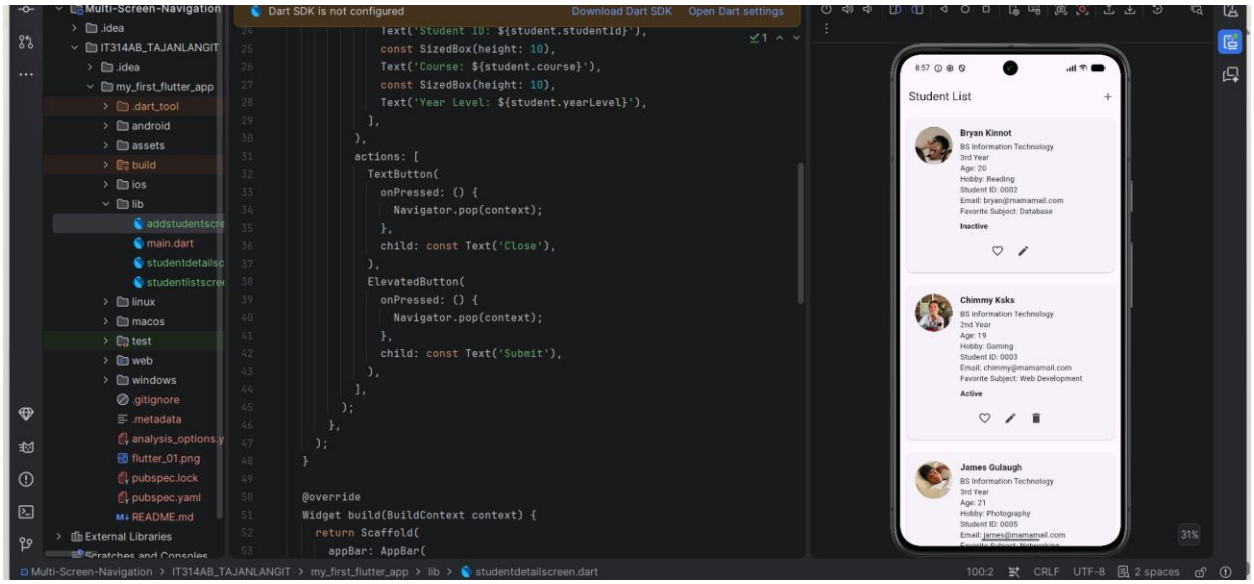
Student List
├── Student Details
│ └── Edit Modal
│
└── Add Student

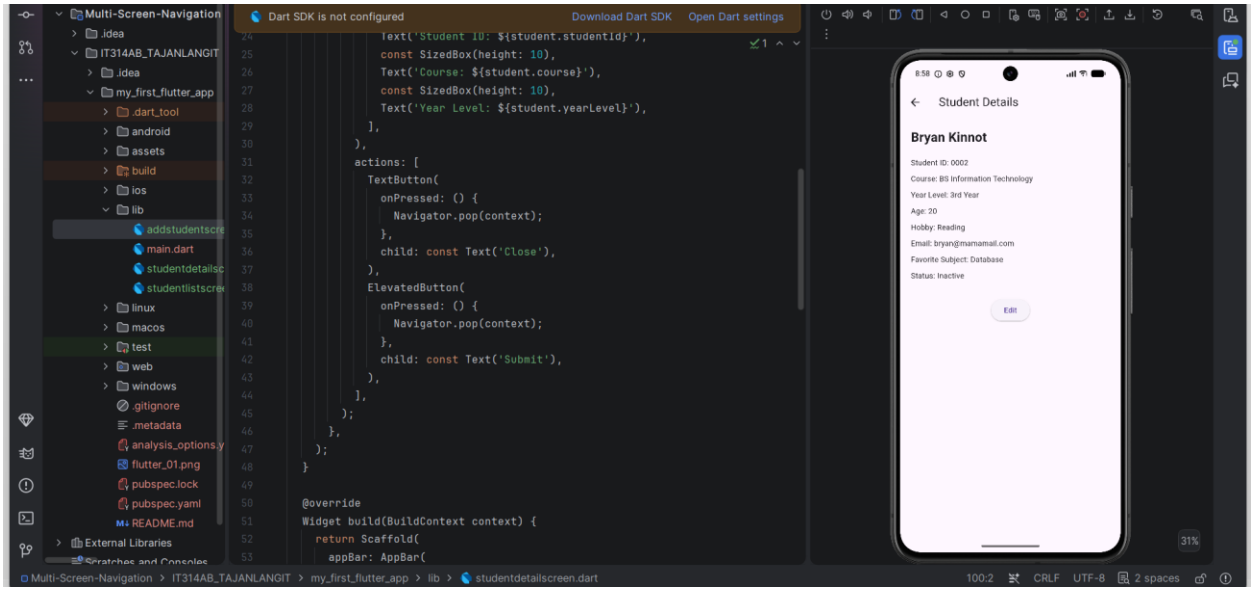
Test

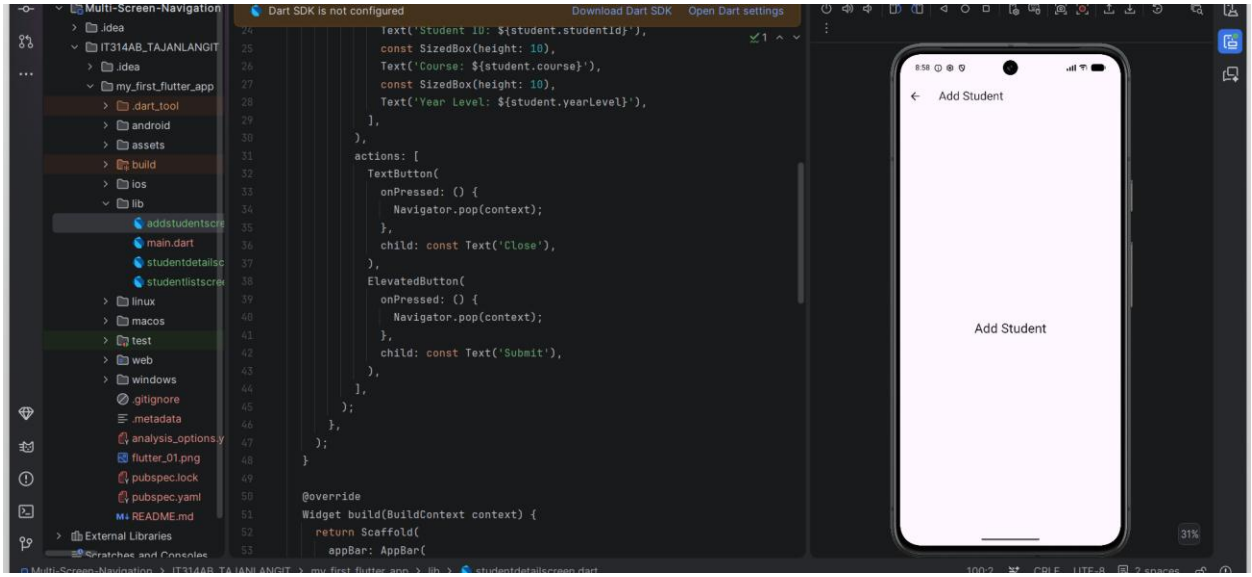
You should be able to navigate between the appropriate screens without placing everything on one screen.

Screenshot:

- Student List
- Student Details
- Edit Modal
- Add Student Screen







🏴 Flag 7 — Connect the Routes

Objective: Organize the application's screens using Flutter routes.

Create routes for the major screens:

/student-list
/student-details
/add-student

Research:

- Named routes
- `Navigator.pushNamed`
- Passing arguments through routes

The **Edit Modal should not be a route**.

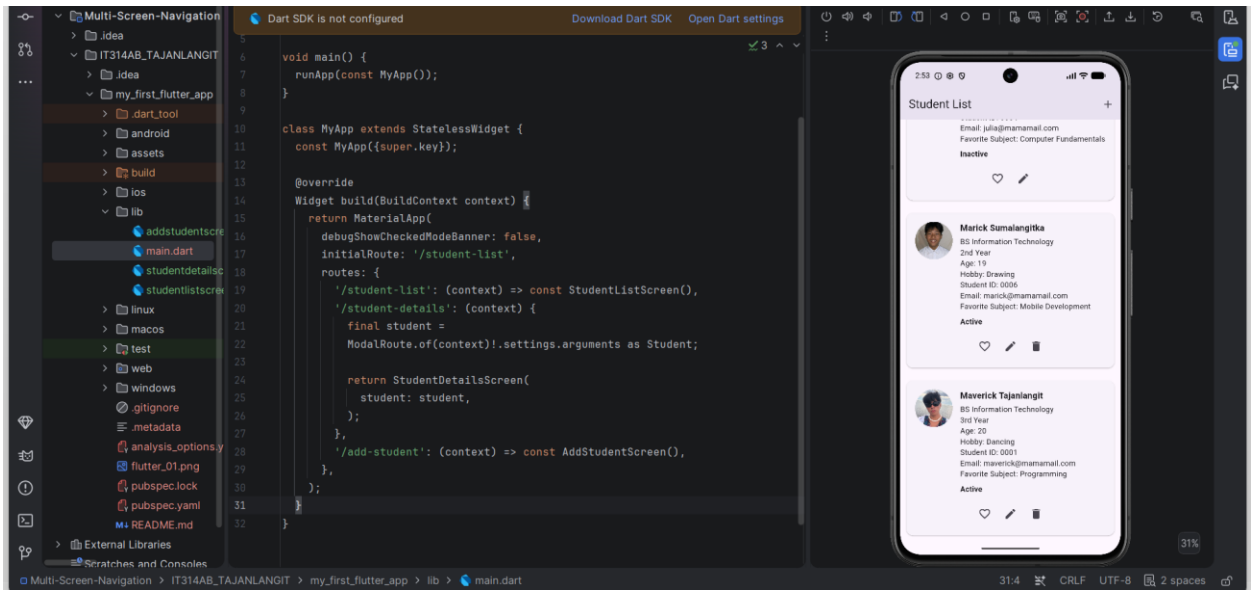
Test

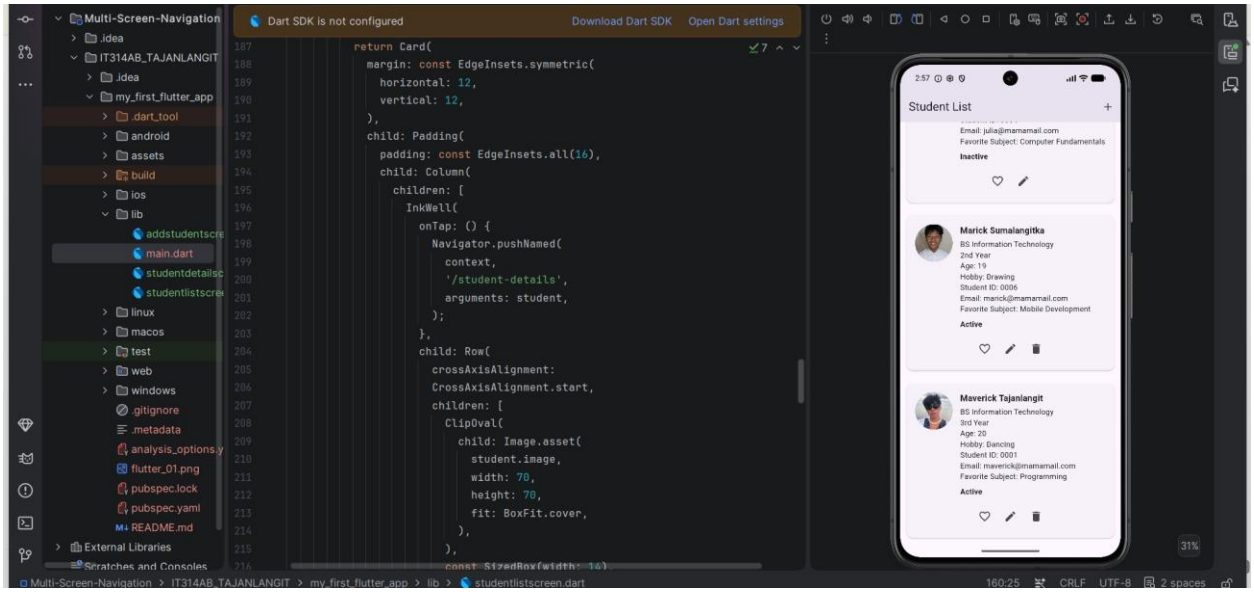
Verify that:

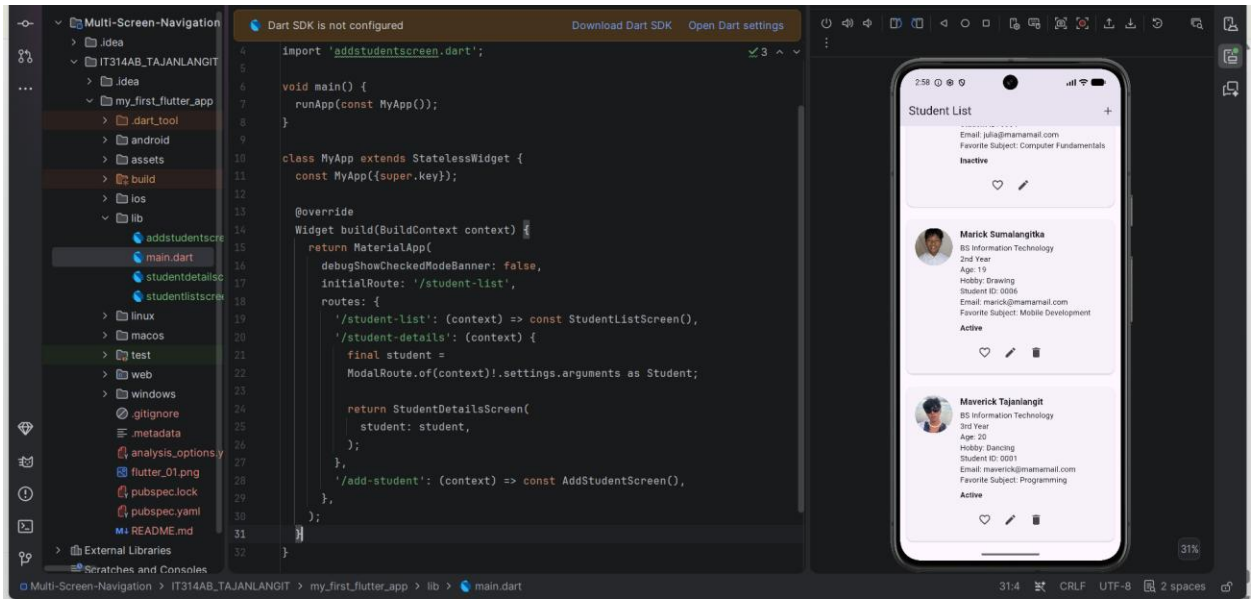
- Student List opens correctly.
- Student Details opens correctly.
- The correct student data is passed.
- Add Student opens correctly.
- Edit opens as a modal from Student Details.

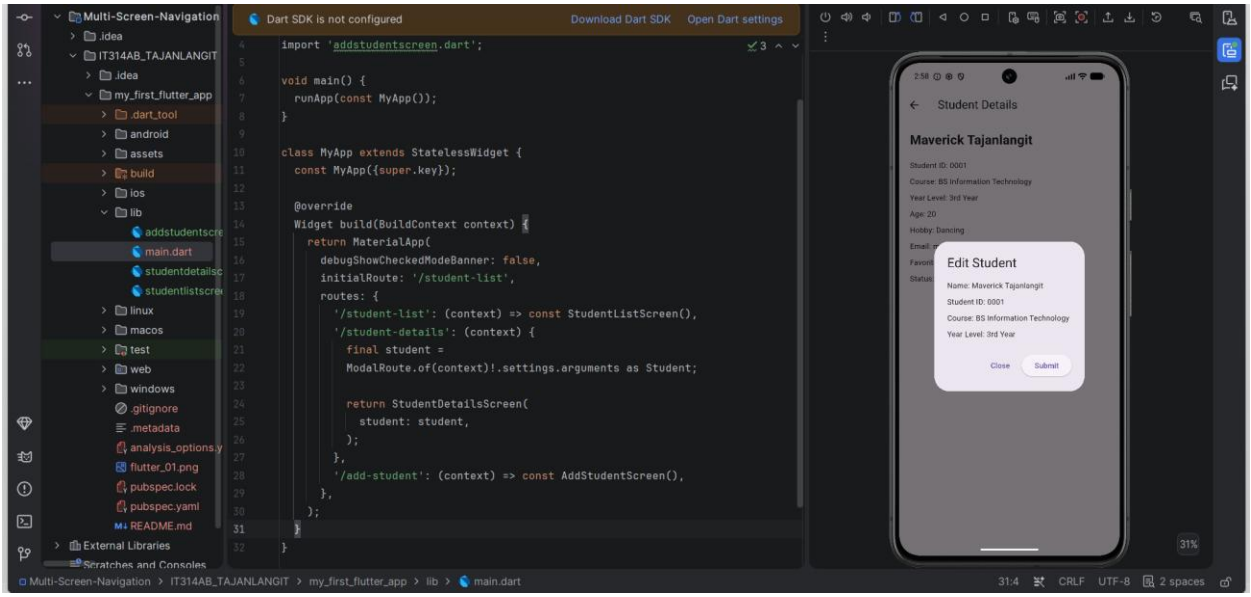
Screenshot:

- Route configuration
- Navigation using routes
- Student Details receiving student data
- Edit modal




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.metadata
analysis_options.y
flutter_01.png
pubspec.lock
pubspec.yaml
README.md
External Libraries
Scratches and Consoles
return Card(
margin: const EdgeInsets.symmetric(
horizontal: 12,
vertical: 12,
),
child: Padding(
padding: const EdgeInsets.all(16),
child: Column(
children: [
InkWell(
onTap: () {
Navigator.pushNamed(
context,
'/student-details',
arguments: student,
);
},
child: Row(
crossAxisAlignment:
CrossAxisAlignment.start,
children: [
ClipOval(
child: Image.asset(
student.image,
width: 70,
height: 70,
fit: BoxFit.cover,
),
),
const SizedBox(width: 14)
Student List
Email: julia@mamamail.com
Favorite Subject: Computer Fundamentals
Inactive
Marick Sumalangitka
BS Information Technology
2nd Year
Age: 19
Hobby: Drawing
Student ID: 0006
Email: marick@mamamail.com
Favorite Subject: Mobile Development
Active
Maverick Tajanlangit
BS Information Technology
3rd Year
Age: 20
Hobby: Dancing
Student ID: 0001
Email: maverick@mamamail.com
Favorite Subject: Programming
Active
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web
windows
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.metadata
analysis_options.y
flutter_01.png
pubspec.lock
pubspec.yaml
README.md
External Libraries
Scratches and Consoles
import 'addstudentscreen.dart';
void main() {
runApp(const MyApp());
}
class MyApp extends StatelessWidget {
const MyApp({super.key});
@override
Widget build(BuildContext context) {
return MaterialApp(
debugShowCheckedModeBanner: false,
initialRoute: '/student-list',
routes: {
'/student-list': (context) => const StudentListScreen(),
'/student-details': (context) {
final student =
ModalRoute.of(context)!.settings.arguments as Student;
return StudentDetailsScreen(
student: student,
);
},
'/add-student': (context) => const AddStudentScreen(),
},
);
}
Student List
Email: julia@mamamail.com
Favorite Subject: Computer Fundamentals
Inactive
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🏴 Flag 8 — Save Your Progress

Objective: Commit and push your completed Multi-Screen Student Directory.

Use a proper Git commit message.

Branch:

`feature/multi-screen-navigation`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
PS C:\Users\admin\Documents\FlutterProjects\Multi-Screen-Navigation> git commit -m "feat: Multi-Screen Navigation"
[feature/multi-screen-navigation (root-commit) 5d8eacb] feat: Multi-Screen Navigation
 154 files changed, 10666 insertions(+)
 create mode 100644 .idea/.gitignore
 create mode 100644 .idea/Multi-Screen-Navigation.iml
 create mode 100644 .idea/caches/deviceStreaming.xml
 create mode 100644 .idea/deviceManager.xml
 create mode 100644 .idea/misc.xml
 create mode 100644 .idea/modules.xml
 create mode 100644 .idea/vcs.xml
 create mode 100644 IT314AB_TAJANLANGIT/.idea/.gitignore
```

```
 create mode 100644 IT314AB_TAJANLANGIT/my_first_flutter_app/windows/runner/win32_window.cpp
 create mode 100644 IT314AB_TAJANLANGIT/my_first_flutter_app/windows/runner/win32_window.h
PS C:\Users\admin\Documents\FlutterProjects\Multi-Screen-Navigation> git branch
* feature/multi-screen-navigation
PS C:\Users\admin\Documents\FlutterProjects\Multi-Screen-Navigation>
```